



Investigation report

Rejected take-off and runway overrun, Tarlton Springs - AAIB-style causal and contributory factors report

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INVESTIGATION WORKFLOW SUITE

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Synopsis

On 18/06/2026 Saltbush Air SB512, a 50-seat twin-turboprop registered VH-SBQ, began its take-off roll on runway 09 at Tarlton Springs Aerodrome at 17:30 local time. This was five minutes after the end of civil twilight, with no moon. Two seconds after V1, the first officer called two kangaroos crossing onto the runway from the northern side. The pilot in command rejected the take-off at 118 kt, six knots above V1. About one second later the No. 2 propeller and right main landing gear struck both animals. The aircraft left the runway end at approximately 21 kt and came to rest upright, 62 m into the 90 m runway end safety area and 28 m short of its far edge. None of the 47 people on board was seriously injured, although one cabin crew member reported bruising during disembarkation. Three No. 2 propeller blades were damaged beyond limits, the right main gear doors were distorted, both right main tyres deflated and the lower right fuselage was punctured. Two kangaroos were killed, and the aerodrome was closed for 4 hours 20 minutes.

The scheduled accelerate-stop distance assumed a stop begun at or before V1. Rejecting after V1 left 695 m for a stop that needed 757 m, which very likely contributed to the overrun. The strike also removed No. 2 engine reverse thrust, and deceleration was approximately 15% below the certified value. The absence of training or guidance for a runway obstruction during the take-off roll likely contributed to the decision to reject. The pilot in command stated, however, that he knew the procedure and departed from it deliberately. Darkness likely contributed by hiding the animals until they reached the lit surface. The operator had approved the evening service without reopening a route risk assessment that restricted Tarlton Springs to daylight operations, and this likely contributed to that darkness.

Several factors explain why the animals were on the runway. The departure fell within the period of most kangaroo activity in the aerodrome's own log. The northern fence had failed and stayed unrepaired through three budget rounds. No runway inspection was requested, and no document had prompted either pilot to consider one. Each of these likely contributed. The operator required an inspection only where a wildlife strike had been reported in the previous 24 months, and that criterion very likely contributed to the omission. The threshold in Aerodrome Standard AS-139 meant the aerodrome needed no documented wildlife plan. This likely contributed, although with less confidence than the other factors.

Two further safety issues did not contribute but increase risk to future operations. They are a roster that closes five-sector duties within 45 minutes of the limit, and floor escape path lighting checked only every 24 months. Maximum braking was established within 2 seconds of the decision, against a fleet mean of 3.2 seconds, and this very likely limited the overrun. A tailwind and an engine power loss before the reject were ruled out. The investigation could not determine whether the crew were fatigued, or whether four brighter-than-nominal edge lights delayed detection of the animals.

The operator suspended the evening service and, from 26/06/2026, listed Tarlton Springs for runway inspections after last light. The aerodrome operator installed a temporary electric wire on 25/06/2026 and programmed fence replacement for 09/2026. One recommendation is made to the operator, to base its inspection listing on wildlife exposure rather than a reported strike.

Factual information

History of the flight

On 18/06/2026 Saltbush Air operated SB512, a scheduled service from Tarlton Springs Aerodrome to Port Carrick, with a 50-seat twin-turboprop aeroplane registered VH-SBQ. It was the last scheduled sector of the day and the fifth of the flight crew's duty. The evening rotation had been added to that duty on the morning of the occurrence, when a member of the rostered crew reported unfit. All times are UTC. The occurrence took place at 17:30 local time.

The aerodrome's only runway inspection of the day was at 21:35 on 17/06/2026, before the first scheduled arrival. It recorded the movement area as clear, and no further inspection was carried out or requested.

A ground delay at Port Carrick earlier in the day was recovered on the turnaround. The doors closed on schedule at 07:16, with 44 passengers, 2 flight crew and 1 cabin crew on board. The occurrence notification states that "The aerodrome was in darkness." Tarlton Springs has no control tower, so the crew made their own CTAF broadcasts. They switched on the pilot-activated runway lighting at 07:21:50, broadcast their taxi at 07:23:15 and broadcast lining up on runway 09 at 07:29:10. Neither the crew nor the aerodrome raised a runway inspection. No wildlife transmission was heard on the CTAF in the preceding 90 minutes. The first officer was pilot flying.

Take-off power was set at 07:30:02. The pilot in command called 80 kt at 07:30:15 and V1 at the calculated 112 kt at 07:30:29. Two seconds later, at 116 kt, the first officer called two kangaroos on the left. The notification records that they were moving onto the runway from the northern side, past the 1,100 m point. The first officer estimated their range at first sight as about 200 m, and said it could have been less. The pilot in command, who stated that after V1 he had been looking at the engine page, did not see the animals before the call.

At 07:30:32 the pilot in command took control and rejected the take-off at 118 kt, six knots and three seconds after V1, with the aircraft 1,152 m along runway 09 and 695 m of sealed surface remaining. About one second later it struck both animals, the impact taken by the right main landing gear and the No. 2 propeller. Maximum braking was established at 07:30:34. Reverse was selected at 07:30:36, but only No. 1 engaged. The aircraft left the sealed surface at 07:30:51 and came to rest at 07:30:57 in the 90 m runway end safety area (RESA), 62 m beyond the runway end. It was upright and on the extended centreline, with the nose wheel in graded gravel. There was no fire.

The pilot in command set the parking brake, shut down both engines, told the cabin to remain seated and asked the aerodrome reporting officer on the CTAF to activate the aerodrome emergency plan. The officer did so at 07:32:10, sent a vehicle, and at 07:32:24 broadcast that the runway was obstructed and the aerodrome closed. No evacuation was commanded and no slides were deployed. Passengers left by the forward airstair, which the CVR records operating from 07:34:40. The notification gives the disembarkation as taking place "4 minutes 8 seconds after the aircraft stopped", while the operator's timeline gives it as running from 07:34:40 to 07:38:48, a period of the same length. [NOT ESTABLISHED: whether the 4 minutes 8 seconds runs from the aircraft stopping or from the start of the disembarkation]

Nobody was seriously injured. One cabin crew member reported bruising during the disembarkation. Two kangaroos were killed. The aerodrome was closed for 4 hours 20 minutes.

Recorded information

At 12:40 the operator's engineering representative removed the FDR and CVR and sent both to the operator's flight data section, which issued report FDA-2026-0417-R1 on 24/06/2026.

From power set until 07:30:32, engine torques (98.4% and 98.5%) stayed within 0.1% of each other. No exceedance, caution or warning was recorded, and fuel flow, ITT and propeller RPM were within normal scatter. Both power levers reached flight idle within 0.6 seconds of the reject. The impact produced a lateral acceleration spike of 0.31 g to the right and a vertical spike of 1.48 g. Brake pressure was 2,180 psi at the impact and 2,940 psi at 07:30:34, with anti-skid active on all four wheels.

Maximum braking was thus established 2 seconds after the decision, against an operator fleet mean of 3.2 seconds for rejected take-offs. No. 2 propeller RPM decayed from 07:30:33, and the No. 2 reverse interlock did not make. Mean deceleration between 07:30:34 and 07:30:44 was 2.6 m/s², approximately 15% below the certified accelerate-stop value for the weight and runway condition, which assumes symmetric reverse.

The FDR recorded 21 kt indicated airspeed and 20 kt groundspeed as the aircraft left the sealed surface. The notification gives approximately 21 kt. In the gravel, deceleration fell to 1.9 m/s². The stopping distance from 118 kt was 757 m. The fuel levers were moved to cut-off at 07:31:26.

The CVR and CTAF recordings agree with this sequence. At 07:21:56 the pilot in command remarked: "Lights are on. That's thirty minutes on the timer." During the stop he said "We're not stopping on it." at 07:30:47, and the first officer said "End's coming up." at 07:30:49.

Aircraft information

VH-SBQ was at 20,800 kg for the take-off, with flap 10 set. [NOT ESTABLISHED: aircraft type, manufacturer, maintenance history and any deferred defects]

The post-occurrence inspection, from 19/06/2026 to 21/06/2026, found:

- three No. 2 propeller blades damaged beyond repairable limits, with leading edge deformation and one blade tip separated, and animal tissue in two blade roots
- distorted right main gear doors, with the gear structure within limits
- both right main tyres deflated through the fuse plugs, consistent with brake energy
- abrasion and two punctures to the lower right fuselage skin, consistent with propeller debris.

The No. 2 engine required a shock-load inspection. Brake energy absorbed was calculated at 84% of the certified maximum.

Meteorology

At 07:28:51 the first officer noted the AWIS wind as 090 degrees at 6 kt. The recorded headwind component was 4 to 6 kt throughout the roll, consistent with the 07:00 routine observation, and no tailwind component was recorded. V1 had been calculated for a dry runway and a 5 kt headwind component. The roll took place about five minutes after last light, with no moon.

Personnel information

The pilot in command had 8,410 hours in total, 2,140 on type, and had been in command for 3 years 1 month. The duty records give his last cyclic simulator session as 11/03/2026, while the interview background describes his most recent check as a recurrent simulator check in February 2026. His last line check was on 02/12/2025. Both were satisfactory, and he had no prior occurrence history.

The first officer had 1,655 hours in total, 405 on type. The duty records give her time with the company as 14 months. The interview background gives 14 months in the role and 17 months with the organisation, having joined in April 2025. Her last cyclic simulator session was on 28/04/2026, and the interview background records a line check in May 2026. She had flown two previous sectors into Tarlton Springs, on 09/01/2026 and 17/02/2026, both in daylight.

Both pilots signed on at 06:30 local. The occurrence came 11 hours 00 minutes into a planned duty of 13 hours 00 minutes, against a maximum of 13 hours 30 minutes under the company's approved fatigue risk management system. Duty and rest in the preceding seven days were within company limits. Rostering data from 14/07/2025 to 18/06/2026 showed the evening rotation on 289 days, always planned as the fourth and fifth sectors of a duty of 12 hours 45 minutes to 13 hours 15 minutes. It was reassigned to a crew already on duty on six of those days, 18/06/2026 being the sixth. At interview both pilots described themselves as tired. The pilot in command said he was not struggling, and the first officer described it as a normal five-sector day. Both pilots' drug and alcohol tests were negative.

Aerodrome information

The runway 09 RESA is 90 m of graded gravel, with ungraded terrain beyond, and the aircraft stopped 28 m short of the end of the prepared area. The runway survey found continuous tyre marks from 1,160 m to the runway end, and carcasses at 1,188 m and 1,221 m from the runway 09 threshold.

Four northern edge lights were above nominal intensity on the night. The pilot in command described the lighting as brighter than he would want, with black on either side. The first officer described it as normal, noting that she had only daylight landings to compare it with.

The aerodrome had no documented wildlife plan. The northern fence had failed and stayed unrepaired through three budget rounds. The wildlife sighting log covered 01/2025 to 06/2026 and held 46 entries, of which:

- 41 recorded kangaroos on or adjacent to the movement area
- 38 of those 41 were timed between one hour before and two hours after last light
- eleven recorded animals on the sealed surface.

The log is a paper register held in the aerodrome office and is not reported to any operator. The aerodrome reporting officer confirmed that no airline had ever asked for it.

Organisational and management information

The operator's Part A 8.14 and its Appendix 8A list the aerodromes where a runway inspection is requested before the first departure after dark. The requirement applies only where a wildlife strike has been reported in the previous 24 months, and Tarlton Springs was not listed. The evening service began in mid-2025, the sector having previously operated in daylight, and was approved without reopening the daylight-only risk assessment. The pilot in command said no briefing on the change reached him. The first officer said that none of her pre-flight sources mentioned animals: the plan, the charts, the aerodrome directory entry and the company aerodrome brief page.

Both pilots stated that every stop or go scenario in their simulator training had involved an engine problem below V1, and none a runway obstruction. Their standard take-off brief gave the captain the decision to reject below V1 and to continue above it. The pilot in command stated that he had seen kangaroos on the strip in the evening several times and mentioned this to ground staff, but had never filed a report.

Tests and research

A borescope inspection of both engines found no pre-existing damage and no evidence of a power anomaly before the impact. The anti-skid system tested serviceable after the occurrence. There was no brake fire and no hydraulic leak.

Other information

During the disembarkation the first officer noticed two unlit patches of the aisle floor proximity strip towards the rear. The emergency lighting was armed and the ceiling and cabin lights were on at the time. Floor escape path lighting is checked every 24 months and is not checked pre-flight. A functional check was carried out on 24/06/2026. [NOT ESTABLISHED: result of the 24/06/2026 functional check]

After the occurrence, the Head of Flying Operations suspended the evening Tarlton Springs service.

Analysis

External Influences

Standard's threshold left the aerodrome with no documented wildlife plan

Aerodrome Standard AS-139 sets out two tiers of wildlife hazard management. Section 10.4.1 covers every certified aerodrome and states that the operator "must take reasonable measures to reduce the risk to aircraft operations presented by wildlife on or in the vicinity of the aerodrome." A documented plan is required only under section 10.4.2, where the aerodrome "is used by aeroplanes having a maximum take-off weight exceeding 27,000 kg, or serves more than 350,000 passenger movements in a calendar year." Where 10.4.2 does not apply, section 10.4.3 provides that the measures "need not be documented."

Tarlton Springs fell below both limbs. Its heaviest scheduled aeroplane, the twin-turboprop used on the Saltbush Air service, has an MTOW of 22,900 kg, and the aerodrome handled 61,400 passenger movements in 2025. It therefore held no wildlife hazard management plan, no hazard assessment and no dispersal programme. Its only documented measures were these two:

- the single inspection before the first arrival under Aerodrome Manual 6.3
- a sighting log that nobody reviewed.

This position was confirmed against the certified aerodrome register and the aerodrome operator's own compliance file. The text of the standard, the traffic figures and the absence of a plan are documentary and not in dispute. Nothing suggested that a plan would have been held had the threshold not excused it. It is virtually certain that the threshold in AS-139 left the aerodrome without a documented wildlife plan. The aerodrome was not, however, in breach of the standard, and the matter is examined here as an influence from outside the aerodrome rather than as a question of compliance.

The question is whether the absence of a plan influenced what followed. The investigation found that knowledge of the hazard existed at the aerodrome. The sighting log for the period 01/2025 to 06/2026 held 46 entries, and 41 of them recorded kangaroos on or adjacent to the movement area. Of those 41, 38 were timed between one hour before and two hours after last light, and eleven recorded animals on the sealed surface of runway 09/27. The log was a paper register kept in the aerodrome office. It was not reported to any operator, and the aerodrome reporting officer confirmed that no airline had ever asked to see it.

The physical defence against that hazard was also known to have failed. Works register item WR-221, raised in 04/2023, recorded four displaced panels and strained wire on the northern boundary fence between the 700 m and 1,300 m runway markers, with animals observed passing through. Replacement was estimated at \$41,000. The item was deferred at the 2023, 2024 and 2025 budget rounds, each time as unfunded and for review at the next cycle. Item WR-263, raised in 02/2026, recorded that the same section was deteriorating further. A site inspection on 23/06/2026 then found a continuous 140 m section with the bottom two wires separated from the posts and a clear animal pad through it at the 1,050 m mark.

The argument for influence rests on how the fence was treated. Without a plan, nothing at the aerodrome owned the wildlife hazard as a hazard. The failed fence therefore sat in the works register as an ordinary maintenance item, competing for council funds, and it was deferred three times with no hazard assessment behind any of those decisions. A documented plan brings three things with it: a hazard assessment, a named owner, and audit against both. Under such a plan the northern fence would have been a recorded wildlife control, and its failure would have had to be answered for rather than carried forward as a deferrable works item.

The argument against influence was also examined, and it carries real weight. Section 10.4.1 still obliged the aerodrome operator to take reasonable measures. A documented plan is not the only means of repairing a fence or of passing a sighting log to an airline. The deferrals were budget decisions, and there was no evidence that a documented plan would have attracted better funding. A plan is a document, whereas the repair required money.

On balance, the investigation found that the absence of a plan likely influenced the outcome. The deciding consideration was that each deferral was made without a hazard assessment in front of the council, and a hazard assessment is precisely what a plan would have supplied. Even so, this conclusion is less secure than the other influences established in this analysis. The contrary argument was not answered in full, because a plan does not of itself fund a fence. The factor is treated as causal on the understanding that its link to the outcome is likely and not settled.

The factor acted through the failed northern fence, which is argued in its own subsection, and through the crossing of the two kangaroos onto the sealed runway. It had no bearing on the operator's criterion for requesting a runway inspection after last light. That criterion is the operator's own rule, which no aerodrome plan could have changed, and it is argued separately.

The AS-139 threshold that excused Tarlton Springs from documenting a wildlife hazard management plan likely contributed to the occurrence, because it left the known failure of the northern fence to be deferred three times as an unfunded works item with no hazard assessment behind it.

Organisational Influences

Failed northern fence left unrepaired through three budget rounds

The aerodrome operator's works register recorded the failure of the northern perimeter fence in 04/2023 as item WR-221. The entry described four displaced panels and strained wire between the 700 m and 1,300 m runway markers. It noted that animals had been seen passing through and recommended replacement at an estimated \$41,000. The item was deferred as unfunded at the 2023, 2024 and 2025 budget rounds. In 02/2026, item WR-263 recorded the same section deteriorating further. On the site visit of 23/06/2026 the investigation found a continuous 140 m section with the bottom two wires separated from the posts and a clear animal pad through it at the 1,050 m mark (Figure 1). Because the condition is set out in the operator's own register and was confirmed on the ground, it is virtually certain that the fence had failed and remained unrepaired.



Figure 1: The northern perimeter fence at the 1,050 m mark, photographed on the site visit: the fallen section and the animal pad through it. Site inspection photograph SV-2026-0417-07, northern boundary at the 1,050 m runway marker, taken 23/06/2026

The pad lies on the northern side of the runway, the same side as the two carcasses. These were recovered at 1,188 m and 1,221 m from the runway 09 threshold, the first of them 138 m beyond the pad. The register had linked this section to animals passing through from the day the item was raised. The failed section was therefore the route by which these animals reached the movement area. Had WR-221 been actioned in 2023, that route would have been closed in 06/2026.

Two matters limit that conclusion. First, WR-221 recommends replacement without stating a specification, and kangaroos can clear a stock fence, so nothing shows that the fence as replaced would have excluded them. Second, the sighting log does not record where any of its 41 kangaroos came onto the aerodrome. The pad shows that these two animals came through the gap, but it does not show that an intact fence would have stopped them.

It is likely that the unrepaired fence allowed the two kangaroos onto the runway, and on that basis it is a causal factor. The finding is qualified by the fact that whether a replacement fence would have kept kangaroos out could not be established.

Evening service approved without reopening the daylight-only risk assessment

The current route risk assessment for Tarlton Springs is RRA-041 issue 3, dated 11/2022. Among its hazards it lists a "single runway with no lighting redundancy" and "no alternate within 200 nm at night", and it carries the restriction "daylight operations only pending review of night lighting and diversion fuel policy." CMR-2025-118, approved by the Head of Flying Operations on 22/05/2025, introduced an evening rotation from 14/07/2025. This was the first scheduled company operation into the aerodrome after last light. The change record assessed it as a

"schedule change only" and recorded "No new hazard identified". It answered "No" to whether a route risk assessment review was required, and it does not refer to the daylight restriction. RRA-041 was not reopened, and no issue 4 exists. The Safety Department's records show no involvement beyond receipt of the approved record. It is virtually certain that the service was approved without the restriction being reviewed.

The process asked whether the change introduced anything new. It did not ask whether the existing assessment still held. Had the restriction been honoured, every company departure from Tarlton Springs would have stayed in daylight. On 18/06/2026 SB512 left on schedule and began its roll at 17:30 local, five minutes after the end of civil twilight. The darkness of the roll therefore followed from the timetable, not from any delay.

The contrary case is that the restriction names night lighting and diversion fuel, not wildlife. A review might have resolved those two items and approved the service with lighting and fuel conditions, leaving the departure after last light unchanged. That is possible, but it is the less likely outcome, because lifting the restriction would have required a night operation at this aerodrome to be assessed on its own terms. The approver has left the company and could not be interviewed. The reasons for the assessment are therefore unexplained, and the report draws no conclusion about them. The change process could not have altered the fleet-wide Appendix 8A criterion, which is discussed separately.

It is likely that approving the evening service without reopening RRA-041 resulted in the roll taking place in darkness, and it is a causal factor.

Evening rotation rostered to close five-sector duties within 45 minutes of the limit

Over 289 days of operation from 14/07/2025 to 18/06/2026, the rostering system shows that the evening rotation always formed the fourth and fifth sectors of a Port Carrick-based duty. Each of these duties signed on between 06:00 and 07:00 and was planned at between 12 hours 45 minutes and 13 hours 15 minutes, against a maximum of 13 hours 30 minutes for those sign-on times. On six days a rostered crew member reported unfit, and the rotation passed to a crew already on duty, extending that crew's day from three sectors to five. 18/06/2026 was the sixth such day. The crew signed on at 06:30 and took off 11 hours into a planned 13 hour duty. This pattern virtually certainly existed.

The investigation then examined whether the roster had any bearing on this occurrence, and found that it was unlikely to have done so. Both crew members described themselves as tired. Yet the recorded data shows power levers at flight idle within 0.6 seconds of the decision and maximum braking within 2 seconds, against a fleet mean of 3.2 seconds, and no step was omitted on the roll. The one decision a long duty could plausibly have affected was not requesting a runway inspection. No document prompted either crew member to consider that decision, so the roster cannot have caused its omission. Whether the crew were fatigued at all is an unresolved matter and is discussed in its own subsection.

The pattern nonetheless stands on its own. Every operation of the rotation has closed a five-sector duty planned to within 45 minutes of the maximum. It has done so at an aerodrome that RRA-041 records as having no alternate within 200 nm at night and limited engineering support. That exposure is sharpest whenever the rotation is reassigned on the morning of operation.

The roster pattern did not contribute to this occurrence. It is likely, however, that addressing it would reduce future risk, and it is therefore reported as a safety issue that increased risk.

Risk Controls

Inspection after last light required only where a wildlife strike was reported in 24 months

The operator's defence against wildlife on the runway after dark was Operations Manual Part A 8.14. At an aerodrome listed in Appendix 8A, paragraph 8.14.2 required the pilot in command to "request a runway inspection from the aerodrome operator prior to the first departure after last light". It also required that the take-off roll not begin until the result had been passed. At every other aerodrome, 8.14.3 stated that "no specific action is required".

The appendix admitted an aerodrome only where "a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months". Nine aerodromes were listed, and Tarlton Springs had never been one of them. The list was last reviewed on 03/03/2025, four months before the evening service began. Because the manual text and the listing are documentary, it is virtually certain that the requirement did not apply to this departure.

The weakness lay in the trigger rather than in the control itself. Part A contained no provision asking a crew to report a sighting, and it offered no route onto the list other than a reported strike. The aerodrome's own sighting log, by contrast, held 46 entries for the period 01/2025 to 06/2026. Of these, 41 recorded kangaroos on or adjacent to the movement area. Thirty-eight of those 41 were timed between one hour before and two hours after last light, and eleven recorded animals on the sealed surface of runway 09/27. The log was a paper register kept in the aerodrome office. It was not reported to any operator, and no airline had ever asked to see it. The information that would have justified listing the aerodrome therefore existed, but the listing criterion could not draw on it. Under 8.14.4, a review of an aerodrome's status follows a strike. The control could therefore be armed only by the outcome it was meant to prevent.

Had the requirement applied at Tarlton Springs, an inspection would have been requested and its result passed before the roll, and the aerodrome's staff disperse animals when they find them. There is one reservation. An inspection completed some minutes before departure does not guarantee a clear runway at the moment of the roll. That reservation limits the confidence in the link, but it does not break it. The departure without an inspection is itself examined under Individual Actions.

The restriction of the inspection requirement to aerodromes with a reported strike very likely contributed to the departure without an inspection, and it is therefore a causal factor.

No training or guidance for a runway obstruction during the take-off roll

Module 4 of TRG-SYL-07 required crews to demonstrate a rejected take-off "for an engine failure, an engine fire and a configuration warning, each initiated below V1". It also required them to demonstrate "the decision to continue for a failure recognised at or after V1". Its four scripted scenarios were all aircraft system events, and none presented an obstruction, an animal or any other external hazard on the runway. The company's training records show that 214 crew completed the module in the 2025 to 2026 cycle. No crew had been given an external obstruction scenario in the five cycles held on file. Operations Manual Part B 4.6 left the same gap. Paragraph 4.6.4 states that "At or above V1 the take-off shall be continued", yet the section offers no guidance on an obstruction at any speed and no amplification for a case in which continuing is itself hazardous. Both pilots stated independently that every stop or go scenario they had been given had involved an engine. The absence of this training and guidance was extremely likely. The only reservation is that scenario records older than five cycles were not available.

The question of influence was argued in both directions. On one reading, the captain had about one second between the first officer's call at 07:30:31 and the reject at 07:30:32. With no rehearsed frame for the decision, he was considering the question for the first time at 116 kt. On the other

reading, the captain stated plainly that he knew the rule and departed from it deliberately, because he judged continuing to be worse. On that view, obstruction training would not have changed what he knew, and the gap is a fleet finding rather than a cause of this reject. The investigation gave weight to the first reading, because an unrehearsed decision made in about a second is not the same decision as one taken against a briefed and practised frame. The second reading nonetheless remains open. This finding explains the circumstances of the decision. It does not excuse it.

The absence of training and procedural guidance for a runway obstruction likely contributed to the decision to reject above V1, and it is a causal factor. The finding is qualified by the captain's own account, which supports a reading in which it did not shape the decision.

Floor escape path lighting checked only every 24 months, not pre-flight

The pre-flight emergency lighting test on this type illuminates the ceiling emergency lights and the exit signs. It neither illuminates nor monitors the floor proximity escape path segments, which are checked only by a dedicated functional test at the maintenance programme's 24-month interval. During the disembarkation, the first officer noticed parts of the aisle floor strip unlit. A functional check on 24/06/2026 then found two of the six segments aft of row 16 on the left side dark on battery power, and traced the fault to a failed common ballast. The last functional test had been 11 months before the occurrence. The defect was therefore undetectable by the crew or a line engineer, and it could have remained so for up to 13 more months. It is virtually certain that this monitoring gap existed.

It is unlikely that the gap, or the defect it concealed, had any bearing on the overrun or its outcome. No evacuation was commanded, and the main cabin lights were on throughout. The passengers left by the forward airstair 4 minutes 8 seconds after the aircraft stopped, without incident. The gap is nonetheless very likely to matter. A fire indication, smoke or a night evacuation over the wing could each turn dark segments in the aft cabin into an obstacle to an orderly exit. Between functional tests, the true state of this lighting on any aircraft of the type is unknown. [NOT ESTABLISHED: whether the failed ballast part is fitted across the fleet and how many aircraft are currently within the 24-month interval.]

The 24-month interval for checking floor escape path lighting is a safety issue reported in its own right, and not a factor in this occurrence.

Local Conditions

Departure in the logged kangaroo peak, 1 h before to 2 h after last light

The aerodrome's wildlife sighting log for 1 January 2025 to 18 June 2026 held 46 entries. Of these, 41 recorded kangaroos on or adjacent to the movement area. Thirty-eight of those 41 were timed between one hour before and two hours after last light, and eleven recorded animals on the sealed surface of runway 09/27. On the day, last light was at 17:25 local, so the window ran from 16:25 to 19:25. The roll began at 17:30. The captain's own experience matched the log. He had seen kangaroos on the strip at Tarlton Springs four or five times, always in the evening and never in daytime.

Two qualifications apply. First, the log is a paper register whose completeness could not be verified. Second, it records sightings rather than presence. Because aerodrome staff go off duty at 18:30, part way through the window, the concentration of entries may partly reflect when somebody was there to see. Neither qualification reverses the pattern. The log places almost all recorded kangaroo movement in this window, and the scheduled departure fell inside it whatever the timetable did.

It was extremely likely that the departure took place in the logged period of peak kangaroo activity. It was likely that this contributed to two kangaroos crossing onto the sealed runway, which places it as a causal factor.

Roll in darkness, 5 minutes after last light, with no moon

The roll began 32 minutes after sunset and 5 minutes after the end of civil twilight, and moonrise was not until 22:41. The only light on the aerodrome came from the runway edge lighting, which the crew had switched on through the pilot-activated lighting. On 23 June 2026, the investigator's site visit at the same time of day confirmed that the unlit graded strip either side of the sealed surface cannot be seen from the centreline. In this light, the first a crew can see of a crossing animal is the animal on the lit surface. The first officer's account on the CVR was consistent with that: "They came out of the dark on the left of the strip" and "There was nothing to see and then there were two of them." In daylight, two animals crossing the strip from the northern fence would have been in view before they reached the seal.

There was evidence the other way. How long the animals had been on the strip before they reached the seal is unknown, and animals that arrived only seconds earlier would have been seen late in daylight too. The captain was heads-in on the engine page after V1, where the procedure puts him. In any light, therefore, the sighting rested on the pilot flying, whose attention belongs on the centreline. These points were outweighed by the site visit and by the first officer's description of animals arriving out of the dark, rather than animals she was slow to see. The unknown time on the strip is the reason the influence is put no higher than likely.

It was virtually certain that the roll took place in full darkness. It was likely that this contributed to the take-off being rejected above V1, which places it as a causal factor. Why the scheduled service rolled after last light is analysed under Organisational Influences.

Four northern edge lights above nominal intensity on the night

A photometric check on 25 June 2026 found four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal intensity for stage 3. This was the section the animals crossed. The lighting runs at one fixed intensity set at the aerodrome's controller, which was found at stage 3, and crews cannot vary it. No work on the lighting was recorded between the occurrence and the check. It is therefore likely that the four lights were at that intensity on the night, although nobody measured them at the time.

Whether that intensity delayed detection could not be settled. The lights stood where the animals crossed. The captain recalled the lighting as "brighter than I'd want", a lit corridor with everything outside it black, and brighter edge lights would deepen the contrast with the dark strip from which the animals came. Against this, the first officer, who saw the animals, described the lighting as normal. She saw them clearly once they were on the seal, at around 200 m, although she said that two daylight landings gave her little basis for comparison. The captain's non-detection is explained by his heads-in duties. Moreover, the strip cannot be seen in darkness whatever the edge lights do, and no evidence links a 131% edge light, rather than the darkness, to the detection of an animal. Settling the matter would need evidence of that link. The question remains unresolved, at about as likely as not, and nothing from it is carried into the conclusions.

Tailwind component during the take-off roll and stop

A tailwind was ruled out as very unlikely. The automatic observations, the recorded FDR wind and the crew's 5 kt headwind performance entry all showed a headwind throughout the roll and the stop.

Flight crew fatigued at the time of the take-off

The take-off was 11 hours into a duty that began at 06:30 local, and it was the fifth sector of a day extended that morning from three sectors to five. Both crew described themselves as tired, and the captain said he would have declined a further sector. Against this, both described the tiredness as ordinary. The captain said "for the two seconds that mattered, I was there". The first officer said it had been a normal five-sector day with a proper break in Port Carrick. The duty was planned at 13 hours against a 13 hour 30 minute maximum, after rest days for both crew. The recorded performance showed no sign of impairment. The power levers reached flight idle within 0.6 seconds, maximum braking followed at 2 seconds against a fleet mean of 3.2 seconds, and the callouts were on time.

Being tired is not the same as being fatigued to a degree that degrades performance. The evidence was evenly balanced, at about as likely as not. Settling it would need a detailed fatigue assessment, including sleep history and the duty pattern before 18 June. It would not, however, alter the analysis. Every mechanism through which fatigue could have acted is contradicted by recorded data, and no document prompted the one decision it might have touched, the request for an inspection. The matter remains unresolved, and nothing from it is carried into the conclusions. The risk in the roster itself is analysed under Organisational Influences.

Individual Actions

No runway inspection requested before departing after last light

Neither the CVR nor the Tarlton Springs CTAF recording contains any request for a runway inspection before departure, any offer of one, or any mention of wildlife. The captain stated that wildlife was not discussed with anyone during the turnaround. The aerodrome's daily inspection log for 18 June 2026 shows a single inspection, at 07:35 local time before the first arrival, which found the movement area clear. No further inspection was carried out or requested that day. That the omission occurred is virtually certain, since two independent recordings, the crew's own accounts and the aerodrome's log all agree.

The inspection was available and would have been carried out. Aerodrome Manual 6.3 provides for an inspection outside the scheduled time when a pilot in command asks for one. The reporting officer was on duty at the terminal during the turnaround and confirmed that an inspection would have been made if requested. SB512 was the only departure after last light that day.

Both crew members said they did not consider an inspection. The captain had expected Tarlton Springs to be on the Appendix 8A list and was surprised to find that it was not. Nothing in the operator's documents prompted either pilot to ask, and the aerodrome had no reason to offer. The omission therefore had no owner. It followed from a control that did not apply at this aerodrome, and that control is examined under Risk Controls.

An inspection would have covered the section from 1,050 m to 1,250 m. The fence pad and both carcasses were later found in that section, and aerodrome staff disperse animals when they find them. Against this, an inspection made some minutes before the roll cannot guarantee that the runway stays clear, and that limitation is why the influence is not put higher. It is likely that the absence of an inspection contributed to the animals being on the runway during the roll, and on that basis the evidence places it as a causal factor.

Take-off rejected 6 kt above V1

The FDR shows V1 called at 112 kt at 07:30:29. Both power levers were retarded to flight idle from 07:30:32 at 118 kt, three seconds and six knots past V1. The CVR records the decision as a single exchange. At 07:30:31 the first officer called "Roos - roos, roos, on the left, they're on the -", and one second later the captain responded "Stopping. I have control." The captain's own account,

given in interview and in his air safety report, corroborates this. That the reject was made above V1 is virtually certain.

Part B 4.6.4 requires the take-off to be continued at or above V1, while section 4.6 gives no guidance on an obstruction seen on the runway, at any speed. In his account, the captain judged that rotating into two large animals would leave the aircraft airborne at night with damage he could not assess, over country offering no usable option. He chose the risk whose end he could see, and he acknowledged that this was not the procedure. His decision was shaped by conditions examined elsewhere in this report. The absence of training or guidance for a runway obstruction is examined under Risk Controls, and the darkness and the presence of the animals under Local Conditions. Whether the edge lighting delayed detection remained unresolved under Local Conditions, and nothing from that matter is carried here.

The late reject affected distance, not deceleration. The scheduled accelerate-stop distance of 1,780 m assumes the stop begins at or before V1, and the margin between the distance required and the distance available was 67 m. The reject began 1,152 m along runway 09, with 695 m of sealed surface left. The stopping distance achieved from 118 kt was 757 m. Beginning the stop three seconds and six knots late therefore consumed the whole margin and more, before the loss of No. 2 reverse is taken into account. It is extremely likely that the reject above V1 contributed to the overrun, and the evidence places it as a causal factor. It is not, however, the origin of the sequence. It sits downstream of the conditions that put the animals on the runway and gave the captain no guidance for meeting them.

Maximum braking established within 2 seconds of the reject decision

The FDR shows that the power levers reached flight idle 0.6 seconds after the decision, and that maximum braking was established at 07:30:34, two seconds after it. The operator's recorded fleet mean from decision to maximum braking is 3.2 seconds. Anti-skid was active on all four wheels throughout, and brake energy reached 84% of the certified maximum without a brake fire. No. 1 reverse was selected at 07:30:36. No. 2 reverse was unavailable, because the strike had destroyed its propeller. The first officer made the speed callouts and the call that the runway end was approaching. That this response occurred is virtually certain, resting on recorded data throughout.

The aircraft left the sealed surface at 21 kt and stopped 62 m into the 90 m RESA. It came to rest upright on the extended centreline and was disembarked without an evacuation. On the investigation's estimate, a stop begun at the fleet mean response would have carried the aircraft a further 30 to 40 m. It is very likely that the speed of the braking response reduced the severity of the outcome, and this was a positive factor. The reject was outside procedure, but the stop that followed was flown faster than the fleet average. The difference was a low-speed stop on a prepared surface rather than a departure from the RESA at speed.

Occurrence Events

Two kangaroos crossed the strip onto the sealed runway

It is virtually certain that two kangaroos entered the runway strip and reached the sealed surface of runway 09 during the take-off roll. Three independent strands of evidence each establish this on their own:

- Two adult carcasses were recovered from the northern half of the runway, at 1,188 m and 1,221 m from the threshold.
- The first officer's sighting call is on the CVR at 07:30:31.
- The FDR recorded the impact two seconds later.

The first officer saw the animals crossing from the northern side towards the south and estimated their range at first sight as around 200 m. The physical layout of the aerodrome is consistent with that account. The animal pad through the failed northern fence lies at the 1,050 m mark, 138 m before the first carcass and on the same side of the runway. The aerodrome's sighting log also shows that kangaroos had been on the sealed surface of runway 09/27 on eleven earlier occasions, all around last light. The route by which the animals reached the strip, and the part played by the fence, are argued under Organisational Influences. What is settled here is only that the animals were on the seal.

That presence extremely likely influenced the occurrence, because both the reject and the strike follow from it directly. The sighting call is the only trigger for the reject on the CVR, and the captain called "Stopping. I have control." one second after it. The impact at 07:30:33 was with these two animals. Without them on the runway there would have been no reason to reject and nothing to strike, so the evidence puts this as a causal factor.

The presence of two kangaroos on the sealed runway during the take-off roll was a causal factor in the overrun, through both the reject it prompted and the strike that followed.

No. 2 propeller and right main gear struck two kangaroos

It is virtually certain that the aircraft struck both animals at 07:30:33, at 114 kt, one second after the reject began. The FDR recorded a lateral acceleration spike of 0.31 g to the right and a vertical spike of 1.48 g. Post-occurrence inspection found three No. 2 propeller blades damaged beyond repairable limits, one blade tip separated, and animal tissue in two blade roots. The right main gear doors were distorted, and propeller debris had punctured the lower right fuselage skin. The recorded data, the physical damage and the recovered tissue all agree.

For the overrun, what mattered was the strike's effect on the stop rather than the damage itself. No. 2 propeller RPM decayed from the moment of impact, and the No. 2 reverse interlock did not make. When reverse was selected at 07:30:36, it came on No. 1 only, and the captain called "Reverse - number two's not coming." The strike therefore took away half the reverse thrust early in a stop that had no margin to spare. Its influence on the outcome is virtually certain, and it acted through the deceleration shortfall discussed below. It is the reason the aircraft left the runway rather than stopping on it.

The strike on the No. 2 propeller and right main gear was a causal factor, because it removed No. 2 reverse thrust during a stop with no margin to absorb that loss.

Deceleration below the scheduled accelerate-stop performance

The scheduled accelerate-stop distance required was 1,780 m against an ASDA of 1,847 m, a margin of 67 m. That figure assumed the stop began at or before V1 with symmetric reverse, and neither assumption held on the night. Distance reconstruction from recorded groundspeed placed the aircraft 1,152 m along the runway at the reject, leaving 695 m of sealed surface. Maximum braking was established, but reverse was available on No. 1 only. The mean deceleration was 2.6 m/s², approximately 15% below the value used in the certified data for this weight and runway condition, and the stopping distance achieved from 118 kt was 757 m. The aircraft left the seal at 07:30:51 at 21 kt and came to rest 62 m beyond the runway end. The recorded data, the reconstruction and the tyre marks on the runway all agree, and the shortfall existed with virtual certainty.

Its influence is also virtually certain. At the certified deceleration from the same point, the reconstruction puts the stop at roughly 660 m, inside the 695 m that remained. At the deceleration actually achieved, 757 m was needed. The overrun therefore had two causes acting on a single stop. The late reject, argued under Individual Actions, shortened the runway left, and the strike lowered the deceleration available over what remained. Neither alone carried the aircraft beyond

the end, but together they did. The prompt braking, also argued under Individual Actions, narrowed the shortfall but could not close it.

The deceleration shortfall was a causal factor, without which the aircraft would have stopped on the sealed surface despite the late start of the stop.

No. 2 engine power loss before the reject decision

A No. 2 engine power loss before the reject, the usual explanation for a high-speed reject, was examined and found very unlikely. The FDR showed both engines at take-off torque within 0.1% of each other, with no caution, warning or exceedance, until the power levers were retarded at 07:30:32. The CVR contains no engine-related call, and the borescope found no pre-existing damage. The No. 2 RPM decay began only at impact, one second later.

Conclusions

The investigation identified the following causal and contributory factors. They should not be read as apportioning blame or liability to any organisation or individual.

A finding stated without a likelihood term was established as fact.

Findings

1. Aerodrome Standard AS-139 section 10.4.2 required a documented wildlife hazard management plan only at an aerodrome used by aeroplanes above 27,000 kg MTOW or serving more than 350,000 passenger movements a year. Tarlton Springs, whose heaviest scheduled aeroplane was 22,900 kg and which handled 61,400 passenger movements in 2025, fell below both limbs. (Safety issue)
2. The aerodrome held no documented wildlife plan, no hazard assessment and no dispersal programme, and it was not in breach of the standard.
3. The northern boundary fence between the 700 m and 1,300 m runway markers was recorded as failed in works register item WR-221 in 04/2023, with animals observed passing through. It was deferred as unfunded at the 2023, 2024 and 2025 budget rounds with no hazard assessment behind any of the deferrals. (Safety issue)
4. On 23/06/2026 the fence had a continuous 140 m failed section, with an animal pad through it at the 1,050 m mark. The pad was 138 m before the first carcass and on the same side of the runway.
5. The evidence did not establish whether the fence, had it been replaced, would have excluded kangaroos.
6. Route risk assessment RRA-041 issue 3 restricted Tarlton Springs to daylight operations. CMR-2025-118, approved on 22/05/2025, introduced the evening service as a schedule change only, and it neither referenced nor reopened that restriction. (Safety issue)
7. The approver of CMR-2025-118 had left the company and could not be interviewed, so what a review of RRA-041 would have concluded could not be determined.
8. Between 14/07/2025 and 18/06/2026 the evening rotation operated on 289 days. Each time it was rostered as the fourth and fifth sectors of a duty planned at between 12 hours 45 minutes and 13 hours 15 minutes, against a maximum of 13 hours 30 minutes. It was reassigned to a crew already on duty on six of those days, and 18/06/2026 was the sixth. (Safety issue)
9. The roster was not found to have contributed to the occurrence, because each mechanism through which it would have had to act was contradicted by recorded data.
10. The operator's Appendix 8A admitted an aerodrome only where a wildlife strike involving a company aircraft had been reported there in the preceding 24 months. No provision asked a crew to report a sighting, and Tarlton Springs had never been listed. (Safety issue)
11. None of the four scripted rejected take-off scenarios in TRG-SYL-07 module 4 presented an obstruction or any other external hazard on the runway, and no such scenario had been presented to any crew in the five training cycles on file. Operations Manual Part B 4.6 contained no guidance on an obstruction seen on the runway at any speed. (Safety issue)
12. The investigation found two readings of how the absence of obstruction training and guidance bore on the decision to reject. On the first, the pilot in command made the decision unrehearsed, in about a second. On the second, he knew the rule and judged continuing to be the worse risk. The evidence supported both readings.

13. The pre-flight emergency lighting test did not illuminate or monitor the floor proximity escape path segments, which were checked only by a functional test at a 24-month interval. On VH-SBQ the functional check on 24/06/2026 found two of the six segments aft of row 16 on the left side dark on battery power, traced to a failed common ballast. (Safety issue)
14. Whether the same ballast part was fitted across the fleet was not established.
15. The floor lighting defect did not affect the overrun or its outcome. No evacuation was commanded, the main cabin lights were on throughout, and the disembarkation was completed without incident.
16. Of the 41 kangaroo entries in the aerodrome's sighting log, 38 fell between one hour before and two hours after last light. On 18/06/2026 that window ran from 16:25 to 19:25 local, and the roll began at 17:30.
17. The log recorded sightings rather than presence. Its completeness could not be verified, and aerodrome staff went off duty at 18:30, part way through the window.
18. The roll began 32 minutes after sunset and 5 minutes after the end of civil twilight, with moonrise not until 22:41. From the centreline the unlit graded strip either side of the sealed surface could not be seen.
19. A photometric check on 25/06/2026 measured four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal intensity for stage 3. The lights were likely at that intensity on the night.
20. Whether that intensity delayed the detection of the animals could not be determined. The pilot in command described the lighting as brighter than he would want. The first officer, who saw the animals, described it as normal and saw them clearly on the sealed surface. Darkness alone explained why nothing beside the seal could be seen. No evidence linking an edge light of that intensity to the detection of an animal was available, and such evidence would have been needed to settle the question.
21. No tailwind component existed at any stage of the roll or the stop.
22. Whether the flight crew were fatigued at the time of the take-off could not be determined. Both pilots described themselves as tired 11 hours into an extended duty. Both also described that tiredness as ordinary and not impairing, and the recorded performance showed none of the usual signs.
23. Settling the question of fatigue would have required a detailed assessment, including sleep history and the duty pattern before 18/06/2026. That assessment was not undertaken because every mechanism through which fatigue could have acted in this occurrence was contradicted by recorded data.
24. No runway inspection was requested or offered before SB512 departed after last light. The reporting officer was on duty and would have carried one out if asked, but no document prompted either pilot to consider it.
25. The pilot in command rejected the take-off at 07:30:32 at 118 kt, three seconds and six knots after V1 at 112 kt, contrary to Part B 4.6.4. At that point the aircraft was 1,152 m along runway 09 with 695 m of sealed surface remaining.
26. The pilot in command judged that continuing into two large animals at night would leave the aircraft airborne with damage that could not be assessed, and he stated that the decision was not by the book.

27. The power levers reached flight idle 0.6 seconds after the reject decision, and maximum braking was established 2 seconds after it, against a fleet mean of 3.2 seconds.

28. Two kangaroos crossed from the northern side onto the sealed runway and were called by the first officer at 07:30:31. Their carcasses were recovered at 1,188 m and 1,221 m from the threshold.

29. At 07:30:33, at 114 kt, the No. 2 propeller and right main gear struck both animals. The strike destroyed the No. 2 propeller's serviceability, and the No. 2 reverse interlock did not make.

30. With reverse on No. 1 only, mean deceleration was 2.6 m/s², approximately 15% below the certified value. The stop needed 757 m, where the certified deceleration from the same point would have needed roughly 660 m.

31. The late start of the stop and the reduced deceleration were each necessary for the aircraft to reach the runway end, and neither alone would have taken it there.

32. No loss of power on the No. 2 engine occurred before the reject decision. The power loss began at the moment of impact, one second after the reject.

Causal factors

- It is virtually certain that the threshold in AS-139 10.4.2 left the aerodrome with no documented wildlife plan. It is likely, though less firmly than most of the factors below, that this contributed to the occurrence. Without a plan, the failed fence was deferred as an ordinary works item with no hazard assessment before the council, although a plan would not by itself have funded the repair. (Safety issue)
- The failed northern fence was left unrepaired through three budget rounds, which is virtually certain. This likely contributed by leaving open the route through which the two kangaroos reached the runway, although whether the fence as replaced would have kept kangaroos out was not established. (Safety issue)
- It is very likely that SB512 departed in the logged kangaroo peak, 1 h before to 2 h after last light. It is likely that this contributed by placing the take-off in the period when animals were most often recorded on the movement area, bearing in mind that the log recorded sightings rather than presence.
- That an inspection after last light was required only where a wildlife strike had been reported in the preceding 24 months is virtually certain. It is very likely that this contributed, because it kept Tarlton Springs off Appendix 8A and so left no requirement for an inspection before SB512's roll. (Safety issue)
- It is virtually certain that no runway inspection was requested before departing after last light. It is likely that this contributed, since an inspection would have covered the section where the animals came onto the runway, although an inspection some minutes before the roll could not have guaranteed a clear runway.
- Two kangaroos crossed the strip onto the sealed runway, which is virtually certain. This very likely contributed, because both the reject and the strike followed directly from the animals being on the surface.
- It is very likely that there was no training or guidance for a runway obstruction during the take-off roll. It is likely that this contributed to the decision to reject above V₁, although the evidence also supported the reading that the pilot in command, who knew the rule, would have judged continuing to be the worse risk in any case. (Safety issue)
- Approval of the evening service without reopening the daylight-only risk assessment is virtually certain. It is likely that this contributed by scheduling SB512's roll after last light,

although what a review would have concluded is inferred, because the approver could not be asked. (Safety issue)

- It is virtually certain that the roll took place in darkness, 5 minutes after last light, with no moon. It is likely that this contributed to the late sighting of the animals and so to the reject, although how long the animals had been on the strip before reaching the seal is not known.
- The take-off was rejected 6 kt above V1, which is virtually certain. This very likely contributed, because beginning the stop at 1,152 m consumed the 67 m scheduled margin, and more, before the loss of reverse was counted.
- It is virtually certain that the No. 2 propeller and right main gear struck two kangaroos. It is virtually certain that this contributed, because it removed the No. 2 reverse one second into a stop with no margin to absorb the loss.
- Deceleration below the scheduled accelerate-stop performance is virtually certain. It is virtually certain that this contributed, because the certified deceleration would have stopped the aircraft in roughly 660 m, within the 695 m remaining, where it needed 757 m.

Contributory factors

- It is virtually certain that the evening rotation was rostered to close five-sector duties within 45 minutes of the limit. It is likely that this increased risk on every operation of the rotation, and most sharply when a crew member reported unfit, although it was not found to have contributed to this occurrence. (Safety issue)
- Floor escape path lighting checked only every 24 months, not pre-flight, is virtually certain. It is very likely that this increased risk, because the state of the escape path lighting on any aircraft of the type was unknown between functional tests, although it did not affect this occurrence or its outcome. (Safety issue)

Other findings

- A tailwind component during the take-off roll and stop was examined and ruled out. Both the automatic weather station observations and the recorded flight data showed a headwind component throughout.
- No. 2 engine power loss before the reject decision was examined and ruled out. Both engines were symmetric until the reject, and the No. 2 power loss began at the moment of impact.
- Maximum braking was established within 2 seconds of the reject decision, faster than the fleet mean of 3.2 seconds. This was examined and found to reduce risk, because it kept the aircraft 28 m short of the end of the prepared runway end safety area.

Safety action and recommendations

Standard's threshold left the aerodrome with no documented wildlife plan

Aerodrome Standard AS-139 requires a documented wildlife hazard management plan only where an aerodrome is used by aeroplanes above 27,000 kg MTOW or serves more than 350,000 passenger movements a year. Tarlton Springs met neither condition. Its heaviest scheduled aeroplane was 22,900 kg, and it handled 61,400 passenger movements in 2025. Under section 10.4.3 its wildlife measures did not need to be documented at all. The aerodrome operator accordingly held no plan, no hazard assessment and no dispersal programme. The aerodrome was not in breach of the standard. The safety issue lies in the threshold itself, which leaves an aerodrome of this size without a documented and audited wildlife programme.

The issue matters because, without a plan, nothing at the aerodrome owned the wildlife hazard as a hazard. It is likely that this allowed the failed northern fence to be treated as an ordinary works item. That item was deferred through three budget rounds without any hazard assessment being put before the council. The investigation therefore treats this issue as a causal factor. That conclusion is held with less confidence than the other causal factors in this report. Section 10.4.1 still required the aerodrome operator to take reasonable measures, and there was no evidence that a documented plan would have secured funding for the repair.

No safety action has been recorded in response to this issue, and this report makes no recommendation on it.

Failed northern fence left unrepaired through three budget rounds

The northern perimeter fence at Tarlton Springs failed between the 700 m and 1,300 m runway markers. The aerodrome operator recorded the failure in its works register as item WR-221 in 04/2023, noting animals passing through. Replacement was deferred as unfunded at the 2023, 2024 and 2025 budget rounds. By 23/06/2026 a continuous 140 m section had failed, with an animal pad through it at the 1,050 m mark (Figure 1).

The failed section was likely the route by which the two kangaroos reached the runway. For more than three years it left open a known path between the paddock and the movement area. The register does not state a specification for the replacement. It was therefore not established whether a replacement fence would have excluded kangaroos, which can clear a stock fence.

In response to the occurrence, the aerodrome operator funded item WR-221 from its 2026 contingency. It programmed replacement of the 140 m northern boundary fence for 09/2026, and it installed a temporary electric wire on 25/06/2026. The action was in progress, and completion of the replacement is [NOT ESTABLISHED: whether the fence replacement was completed as programmed in 09/2026].

This action answers the safety issue in part. It closes the failed 140 m section, but WR-221 described a failure extending from the 700 m to the 1,300 m marker. Nothing shows that the replacement fence will be of a specification that excludes kangaroos.

No safety recommendation has been recorded for this safety issue.

Evening service approved without reopening the daylight-only risk assessment

The operator introduced its scheduled evening rotation to Tarlton Springs through CMR-2025-118. The change record assessed this as a schedule change only and concluded that no review of the route risk assessment was required. RRA-041 issue 3, the current assessment for the aerodrome, restricted company operations there to daylight pending review of night lighting and diversion fuel policy. The change was approved without that restriction being referred to or reopened. The process asked whether the change introduced anything new, rather than whether the existing assessment still held. It is virtually certain that the evening service was approved in this way.

The issue matters because the restriction, had it been honoured, would have kept every company departure from Tarlton Springs in daylight. SB512 left on schedule on 18/06/2026 and began its take-off roll at 17:30, after the end of civil twilight at 17:25. It is likely that the approval without review contributed to the roll taking place in darkness.

That finding carries a qualification. The restriction named night lighting and diversion fuel, not wildlife. A review might therefore have approved the evening service with conditions on those two matters, leaving the departure after last light unchanged. The approver has left the company and could not be asked what was considered, so what a review would have concluded is inference. This was judged the less likely of the two outcomes.

Safety action taken

No safety action has been recorded in response to this safety issue.

Safety recommendations

No safety recommendation is made in respect of this safety issue.

Inspection after last light required only where a wildlife strike was reported in 24 months

The operator's Operations Manual Part A required a runway inspection before the first departure after last light only at aerodromes listed in Appendix 8A. An aerodrome qualified for that list only where a wildlife strike involving a company aircraft had been reported there in the preceding 24 months. No provision asked a crew to report a sighting, and none admitted an aerodrome to the list on any other evidence. Tarlton Springs had never been listed, although the aerodrome's own sighting log recorded eleven occasions on which kangaroos were seen on the sealed surface. The control itself was sound, but it could be armed only by the occurrence it existed to prevent. Its absence at Tarlton Springs very likely contributed to the departure taking place without a runway inspection.

Safety action taken

On 26/06/2026 the operator amended Appendix 8A by temporary revision TR-2026-04 to list Tarlton Springs. The 8.14.2 inspection requirement therefore applies to every departure there after last light until a permanent change is made. This action addresses the issue in part. It closes the gap at Tarlton Springs, but the listing criterion is unchanged, so no other aerodrome can be listed before a strike has occurred there.

Safety recommendation

It is recommended that the operator replace the Appendix 8A listing criterion so that an aerodrome is listed on its exposure to wildlife rather than on a reported strike. It is also recommended that the operator introduce a route risk assessment trigger for any aerodrome first operated into after last light.

No training or guidance for a runway obstruction during the take-off roll

The operator's recurrent training and its operating procedures both left out an obstruction on the runway during the take-off roll. Module 4 of TRG-SYL-07 presented four rejected take-off scenarios, and all of them were aircraft system events. No crew had been given an external obstruction scenario in the five training cycles held on file. The current cycle alone covered 214 crew. Operations Manual Part B 4.6 gave no guidance on an obstruction seen on the runway at any speed. Nor did it amplify paragraph 4.6.4 for a case in which continuing the take-off is itself hazardous. It was extremely likely that this gap existed.

The gap matters because it likely contributed to the decision to reject the take-off 6 kt above V1. The captain had about a second to decide, with no briefed or practised frame to decide against. That reading carries a qualification. The captain stated that he knew the rule and departed from it deliberately because he judged the alternative worse. On that account the gap did not shape his decision. On either reading, the gap is not confined to this crew. It applies to every crew trained under the same syllabus and operating to the same manual, and both pilots raised it independently in almost the same words.

No safety action has been recorded in response to this safety issue. [NOT ESTABLISHED: whether the operator has taken or proposed any action on its training syllabus or Operations Manual Part B 4.6.] No safety recommendation has been recorded for this safety issue, and none is made here.

Evening rotation rostered to close five-sector duties within 45 minutes of the limit

The evening rotation between Port Carrick and Tarlton Springs, operated as SB511 and SB512, was rostered so that it closed long five-sector duties close to the maximum permitted. On each of the 289 days it operated from 14/07/2025 to 18/06/2026, it formed the fourth and fifth sectors of a Port Carrick-based duty signing on between 06:00 and 07:00. Those duties were planned at between 12 hours 45 minutes and 13 hours 15 minutes, against a maximum of 13 hours 30 minutes for five sectors from those sign-on times. The roster pattern virtually certainly existed.

It is unlikely that this pattern had any part in the occurrence. It is reported here because it is a standing exposure rather than a feature of one day. It is present on every operation of the rotation. It also places the last sector of the longest duty in the pattern at an aerodrome that RRA-041 records as having no alternate within 200 nm at night and limited engineering support.

The exposure is sharpest when a rostered crew member reports unfit. The rotation then passes to a crew already on duty. On six days in the period, that extended a crew's day from three sectors to five on the morning of operation, and 18/06/2026 was the sixth such day. Addressing the roster pattern would likely reduce the risk to future operations of the rotation.

No safety action has been recorded in response to this issue, and no safety recommendation is made.

Floor escape path lighting checked only every 24 months, not pre-flight

The pre-flight emergency lighting test on this type illuminates the ceiling emergency lights and the exit signs. It does not illuminate or monitor the floor proximity escape path segments. Their only check is a dedicated functional test at the maintenance programme's 24-month interval. Between those tests, the state of the escape path lighting on any aircraft of the type is unknown, and no

check available to a crew or a line engineer will reveal a defect. On VH-SBQ, this gap concealed two dark segments in the aft cabin. They had gone undetected since a functional test 11 months before the occurrence.

The issue had no bearing on the overrun or its outcome. No evacuation was commanded, the main cabin lights were on throughout, and the passengers disembarked by the forward airstair without incident. Even so, it is very likely that the gap matters for safety. A fire indication, smoke or a night evacuation over the wing would change the circumstances. Dark segments in the aft cabin could then be the difference between an orderly exit and a blocked one. The concern is not the defect found on VH-SBQ, which was traced to a failed common ballast. It is the standing absence of monitoring that allowed that defect to remain unseen. [NOT ESTABLISHED: whether the failed ballast part is fitted across the fleet, and how many aircraft of the type are currently within the 24-month interval since their last functional test]

No safety action has been recorded in response to this safety issue. The evidence available does not support a specific safety recommendation, and none is made.

Distribution

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